

## Method Proposal:

### A Multi-Agent Evidence Fusion Framework for Cross-Domain Misinformation Detection Using Affective Retrieval and External Evidence Verification

## Background

The spread of misinformation across domains such as healthcare, finance, politics, and public policy has created a need for reliable automated detection systems. Existing approaches primarily rely on textual and semantic analysis, which often fail to generalize across domains due to differences in vocabulary, writing style, and contextual information. Recent retrieval-augmented Large Language Models (LLMs), particularly emotion-aware frameworks such as RAEmoLLM, have demonstrated that affective information transfers more effectively across domains than semantic similarity alone. However, current systems still depend mainly on retrieved dataset examples and rarely incorporate real-time external evidence or structured reasoning, limiting their effectiveness on emerging news events.

## Research Gap

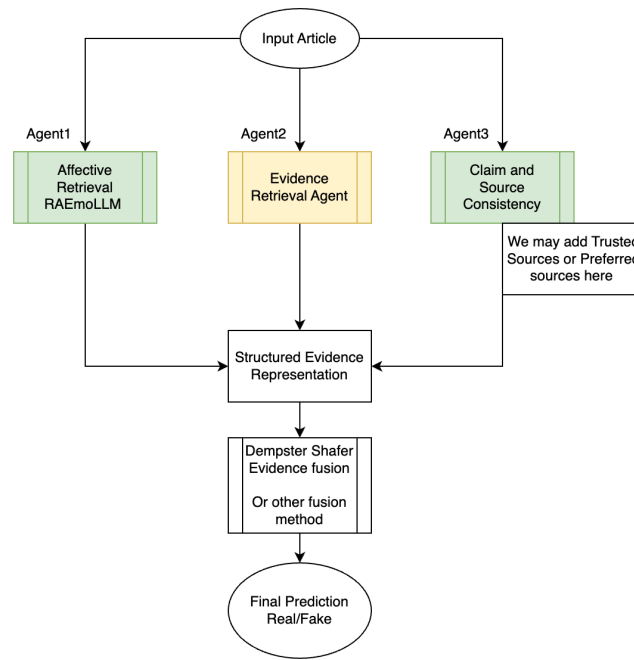
Although affect-aware retrieval improves cross-domain misinformation detection, existing methods largely ignore complementary evidence such as external news verification, claim consistency, and structured factual analysis. Furthermore, most approaches rely on a single decision-making model, reducing transparency and making it difficult to explain how different evidence sources contribute to the final prediction. There is a need for a modular, explainable framework capable of integrating multiple heterogeneous evidence sources while explicitly handling uncertainty.

## Proposed Research

This research proposes a **Multi-Agent Evidence Fusion Framework** that extends the RAEmoLLM retrieval mechanism by incorporating three specialized agents:

1. **Affective Retrieval Agent** -Retrieves emotionally and semantically similar misinformation examples using the RAEmoLLM framework and VREG affective embeddings.
2. **External Evidence Verification Agent** - Retrieves and summarizes supporting or contradictory evidence from trusted online news sources using a web search API, producing an evidence consistency score.
3. **Claim Consistency Agent** - Extracts structured factual information (claims, entities, dates, organizations, and sources) from both the input article and retrieved evidence to assess factual consistency.

The outputs of these agents will be fused using **Dempster–Shafer Evidence Theory (DST)**, enabling mathematically grounded evidence combination while explicitly modeling uncertainty and conflicts among evidence sources. The framework will generate both a final misinformation classification (Real/Fake) and an interpretable confidence score supported by evidence from each agent.



## Objectives

- Develop a modular multi-agent framework for cross-domain misinformation detection.
- Integrate affect-aware retrieval with real-time external evidence verification.
- Apply Dempster–Shafer Evidence Theory for transparent evidence fusion.
- Improve explainability by providing interpretable evidence supporting each prediction.
- Evaluate the proposed framework

## Stack Expected to use:

**Models:** Mistral / Llama / Qwen Family models for Tools calling and Agentic Support.

**Tools:** Tavily API, Serp API, Langchain (for LLM Loading), Langgraph for Agent system.